

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8" />
5    <title>Lab 7.1 D3 Line and Area Chart</title>
6    <script src="https://d3js.org/d3.v7.min.js"></script>
7    <style>
8      .line {
9        fill: none;
10       stroke: steelblue;
11       stroke-width: 2;
12     }
13     .area {
14       fill: lightsteelblue;
15       opacity: 0.5;
16     }
17     .axis path,
18     .axis line {
19       fill: none;
20       stroke: #000;
21       shape-rendering: crispEdges;
22     }
23     .annotation {
24       stroke: red;
25       stroke-dasharray: 5 5;
26     }
27   </style>
28 </head>
29 <body>
30   <h2>Australian Unemployment (1978-1995)</h2>
31   <svg width="700" height="400"></svg>
32
33   <script>
34     const w = 700;
35     const h = 400;
36     const padding = 50;
37
38     const svg = d3.select("svg");
39
40     // Read and parse CSV
41     d3.csv("Unemployment_78-95.csv", function(d) {
42       return {
43         date: new Date(+d.year, +d.month - 1), // month - 1 for JS
44         number: +d.number.replace(/,/g, "") // remove commas from number
45       };
46     }).then(function(data) {
47       // Scales
48       const xScale = d3.scaleTime()
49         .domain(d3.extent(data, d => d.date))
50         .range([padding, w - padding]);
51
52       const yScale = d3.scaleLinear()
53         .domain([0, d3.max(data, d => d.number)])
54         .range([h - padding, padding]);
55
56       // Axes
57       const xAxis = d3.axisBottom(xScale).ticks(10);
58       const yAxis = d3.axisLeft(yScale).ticks(10).tickFormat(d3.format(","));
59
60       svg.append("g")
61         .attr("transform", `translate(0,${h - padding})`)
62         .call(xAxis);
63
64       svg.append("g")
65         .attr("transform", `translate(${padding},0)`)
66         .call(yAxis);
67
68       // Area generator
69       const area = d3.area()
70         .x(d => xScale(d.date))
71         .y0(yScale(0))
72         .y1(d => yScale(d.number));
73
74       // Line generator
75       const line = d3.line()
76         .x(d => xScale(d.date))
77         .y(d => yScale(d.number));
78
79       // Draw Area
80       svg.append("path")
81         .datum(data)
82         .attr("class", "area")
83         .attr("d", area);
84
85       // Draw Line
86       svg.append("path")
87         .datum(data)
88         .attr("class", "line")
89         .attr("d", line);
90
91       // Annotation Line at 500,000
92       svg.append("line")
93         .attr("class", "annotation")
94         .attr("x1", padding)
95         .attr("x2", w - padding)
96         .attr("y1", yScale(500000))
97         .attr("y2", yScale(500000));
98
99       svg.append("text")
100        .attr("x", w - padding)
101        .attr("y", yScale(500000) - 5)
102        .attr("text-anchor", "end")
103        .attr("fill", "red")
104        .text("500,000 unemployed");
105     });
106   </script>
107 </body>
108 </html>

```